HOLD-ON

Project Plan for the Great Indian AI Internship Challenge

You live your life. Hold-On follows up.

> **THE THESIS**
>
> The call is only the beginning. Hold-On turns a customer-care conversation into an evidence-linked case: it records who promised what, when it is due, whether the outcome was verified, and what authorised step should happen next.

Executive summary

Hold-On is a multilingual voice-first consumer advocate for routine service complaints. For the hackathon, it will demonstrate one complete telecom billing dispute using a fictional IVR and a simulated support agent. Prisma transcribes the conversation; an extraction layer proposes ticket details and commitments; deterministic validators and user confirmation control what enters the Promise Ledger; Timbre speaks short replies; and a deadline worker triggers a permissioned mock callback when a case becomes overdue.

The product is not positioned as the first AI that makes calls or waits on hold. Products such as hiRobin already publicly describe outbound calls, customer-support/refund handling and follow-ups. Hold-On will demonstrate a narrower, verifiable workflow: speaker-attributed commitments linked to exact transcript evidence, explicit outcome verification, safe deadline scheduling and a source-backed escalation path. This is a product hypothesis to validate, not a claim that competitors lack every one of these features.

Build decisions at a glance

Decision	Plan
Challenge / deadline	Great Indian AI Internship Challenge; final deadline 10 Nov 2026, 11:59 PM IST. Internal target: submit 9 Nov.
Team	Two people. Assign named owners before coding; each participant is interviewed individually.
Primary category	Real Deal. Support with tested code-mixed speech, noisy telephony and a polished 60-second demo.
MVP scope	One fictional telecom billing complaint, one mock IVR, one simulated agent, one promise ledger and one controlled follow-up loop.
Core models	Prisma v3.0 and Timbre v3.0 APIs (Gnani). Extraction uses Sarvam-30B (open weights, Apache 2.0, no access request), self-hosted on a cloud VM we rent (any provider) to run and test it.
Non-negotiable principle	Model proposes; validation checks; user confirms or authorises; the system records and can explain why.

Document status: Revised working plan. Targets below are pre-build acceptance goals, not measured results. Public competitor and official-process sources were checked on 30 October 2026.

01 / Problem and competitive position

Be precise about the market: calling and follow-up exist already; Hold-On must prove its evidence-linked case workflow is better.

The user problem

A customer reports a wrong bill, a missing refund or an unresolved service request. The first call may produce a ticket number and a verbal commitment, but the customer then has to remember the details, wait for the promised date, repeat the story to another agent and figure out the correct escalation channel. Language barriers and long phone menus make the process harder.

Pain point	Product response
Lost context between calls	A persistent case timeline stores ticket ID, issue, previous attempts and status.
Vague or forgotten commitments	The Promise Ledger records exact wording, speaker, timestamp, normalised due date and confidence.
An AI may mishear dates or amounts	User confirmation is required for ambiguous or material fields; uncertain items stay unconfirmed.
A call ending is mistaken for resolution	Promises have separate states; fulfilment requires user confirmation or an explicit demo-system check.
Consumer is unsure what to do next	A source-linked escalation guide shows the next appropriate channel and drafts, but does not submit, a complaint without approval.

Competitive landscape: honest positioning

Product / category	What is publicly documented	Implication for Hold-On
hiRobin (India)	Its official site says Robin can make calls, sit through hold music, talk to support, handle customer support/refunds, and handle complaints/follow-ups that are “logged and chased until they’re done.”	Direct overlap with basic calling and follow-up. Do not claim this is unique. Benchmark whether Hold-On’s evidence-linked, speaker-attributed commitment records and verification workflow create a demonstrable distinction.
General-purpose AI calling agents	Products in this category make or handle calls for users. Individual features vary and require product-by-product verification.	The category itself is not new. Compete on transparent case state, grounded evidence, safe automation and measurable accuracy.
Official grievance portals / channels	TRAI describes a two-tier telecom complaint route through the provider complaint centre and provider Appellate Authority. NCH is a consumer grievance/pre-litigation channel; CPGRAMS covers eligible public-authority service-delivery grievances.	These are complementary channels, not competitors to claim as absent. Hold-On should guide users to the right route and preserve case evidence.

DEFENSIBLE POSITIONING

“Hold-On links each customer-care commitment to its source transcript, tracks the deadline, distinguishes a stated promise from a verified outcome, and uses that history to guide the next authorised action.” Validate this claim against live products before making any broader exclusivity statement.

Success is not a claim of uniqueness

The competitive gap is a hypothesis to test, not a proven fact. Before submission, compare official demos/listings against a fixed checklist: outbound call, IVR navigation, promise extraction, speaker attribution, quoted source evidence, outcome verification, deadline tracking, duplicate-safe follow-up and escalation workflow. Mark “not publicly documented” where evidence is unavailable; do not infer that a product lacks a feature.

02 / The product: Promise Ledger + verified follow-through

The ledger is the differentiator. Other features support it rather than expanding the MVP into a general platform.

End-to-end user journey

1. User describes a telecom billing issue by voice or text, selects a language, and approves what information the agent may share.
2. Hold-On navigates a fictional IVR and exchanges turns with a simulated customer-care agent.
3. The system proposes ticket details and commitments with speaker role, timestamps and exact transcript excerpts.
4. The user confirms material fields (especially ticket ID, amounts and dates). Low-confidence or ambiguous items cannot become actionable promises.
5. A confirmed promise is stored with its deadline and verification method. A scheduler tracks it; the demo clock can simulate time passing.
6. If overdue and still unresolved, an authorised, duplicate-safe mock callback refers to the earlier ticket and commitment.
7. The user gets an evidence pack and a clearly sourced next step. The case is not marked fulfilled until outcome verification succeeds.

Promise Ledger: required fields

Field group	Required fields / rule
Identity	case_id, promise_id, source_call_id; stable IDs; created_at and updated_at timestamps.
Commitment	commitment_type (refund, correction, callback, technician visit, update), summary, amount/currency if applicable, exact quoted wording.
Attribution	speaker_role (agent, user, IVR, system), speaker confidence, transcript start/end timestamps, and excerpt ID. In the mock, role is assigned from the scripted turn; real-call diarisation must be validated separately.
Timing	spoken deadline phrase, normalized due_at, timezone Asia/Kolkata, deadline type (calendar/working days), parser confidence, user confirmation and confirmation timestamp.
Evidence and state	transcript excerpt, source reference, extraction confidence, status, verification method, verifier, verified_at and verification result.
Automation controls	follow_up_allowed, approval timestamp, attempt_count, next_attempt_at, last_attempt_at, cancellation reason and idempotency key.

Five statuses only

Status	Definition
Pending	Commitment was captured but the due time has not arrived, or confirmation/verification is incomplete.
Due soon	The confirmed deadline is within the configured reminder window.
Overdue	Deadline has passed; outcome remains unverified or unresolved.
Fulfilled	Outcome verified by user confirmation or an explicit test-system check.
Unable to verify	Evidence is insufficient or contradictory; no automated claim of success is made.

> **EVIDENCE RULE**
> A transcript proves only what the system heard or recorded; it is not, by itself, proof of a company's legal obligation. Store the exact excerpt and distinguish "agent stated" from "outcome verified."

03 / Architecture and control flow

A simple, auditable pipeline. Use deterministic code for state, dates and callbacks; use the model for language understanding and phrasing.

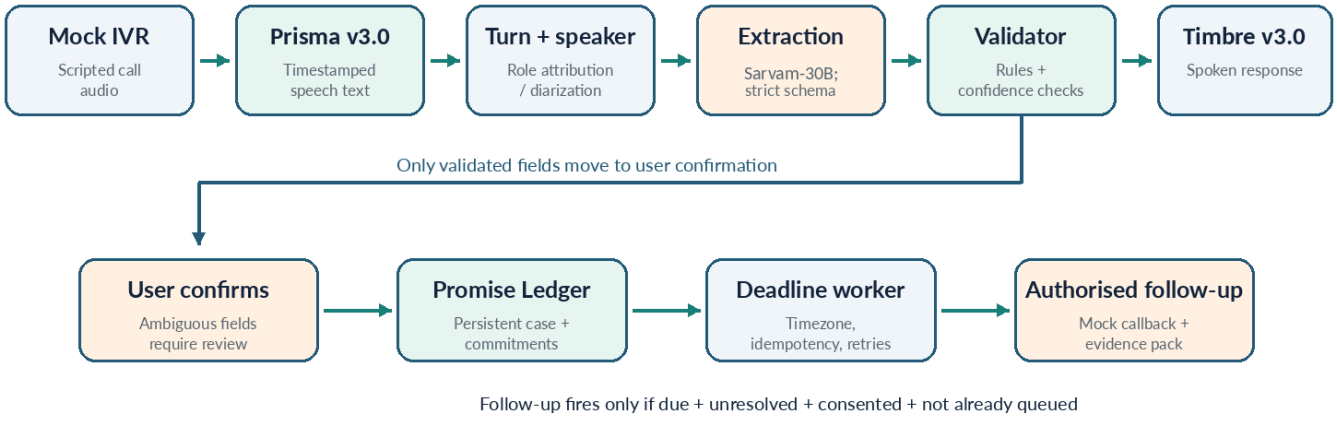


Figure 1. Hackathon architecture. The follow-up path is simulated in the controlled mock IVR; production telephony is out of scope for the MVP.

Control sequence

Component	Responsibility	Must not do
Mock IVR / audio adapter	Play scripted menu/hold audio, route turns to a simulated agent, vary menu branches and inject controlled noise.	No real company helplines or real customer data in the demo.
Prisma adapter	Stream/transcribe supported audio and retain timestamps or chunk metadata where available.	Do not infer that the raw transcript is perfectly accurate.
Turn and speaker layer	In the mock, source turn metadata labels agent/user/IVR; attach timestamps to each utterance. Real deployment requires tested speaker attribution.	Do not assign a financial promise to the wrong speaker silently.
Extractor (Sarvam-30B)	Propose structured fields and next-action candidates from the recent conversation.	Cannot persist promises, schedule callbacks or authorise sharing directly.
Validator + state machine	Schema validation, confidence and ambiguity rules, amount/date checks, valid state transitions, permission checks.	Reject malformed, contradictory or unsupported records.
Timbre adapter	Speak short replies and confirmed summaries; support a user correction or hand-off.	Never speak/store OTPs, PINs, passwords or full card details.
Ledger + deadline worker	Persist case state, schedule due checks, prevent duplicate jobs, maintain audit log and trigger allowed mock follow-ups.	A completed call never automatically marks a commitment fulfilled.

Implementation stack and data storage

Use a React + Vite interface and a Node/Express backend if those are already familiar to the team. Pick one persistence layer on Day 1 and keep it; avoid an infrastructure change mid-sprint. The chosen store must persist cases, commitments, scheduler jobs and callback attempts across server restarts. For a single-machine demo, SQLite is a low-setup option; use MongoDB only if the team already has a stable working connection. Keep model API keys server-side in environment variables.

04 / Model roles, storage and deadline engine

The product must work even if the Sarvam-30B VM setup is delayed. Keep the model behind one OpenAI-compatible URL so a fallback model can be swapped in.

Model plan: Gnani speech models and the Sarvam-30B extractor

Model	MVP role	Day-1 validation / fallback
Prisma v3.0	Speech-to-text for mock IVR and simulated agent speech, including Hindi/Hinglish; retain timestamps/chunk references supported by the API.	Confirm API key and v3.0 access, streaming behaviour, supported audio encoding/sample rate, end-of-utterance handling and latency using the official quick start.
Timbre v3.0	Text-to-speech for concise prompts, confirmation and spoken status updates.	Confirm supported voice/language and streaming interface. If a needed language is unavailable, keep an accessible on-screen text fallback.
Sarvam-30B	Propose commitment fields, dialogue state or next action from a bounded transcript window.	Open weights, Apache 2.0, no access request; self-hosted on a rented VM (see box below), not via Sarvam's deprecated hosted API. Day 1: run 30 to 40 labelled Hinglish transcripts and count invented quotes and wrong dates. If setup is delayed or quality falls short, use another open model (for example Qwen3.5-35B-A3B) on the same VM behind the same schema, or a deterministic rule fallback. Name what was actually used.

WHERE SARVAM-30B RUNS

Sarvam-30B runs on a cloud VM we rent (any provider; vLLM on an 80GB-class GPU, or a GGUF build with llama.cpp on a 48GB-class GPU) for running and testing the model. The backend calls it through one OpenAI-compatible URL in an environment variable. We start the VM for the day-1 test, evaluation runs and the demo recording, and destroy it when idle; laptops use a small local model for development only.

Date and number handling

- Preserve the raw phrase exactly (for example, “paanch working days”) alongside the normalized date; never overwrite the original wording.
- Normalize dates in Asia/Kolkata. Treat “working days” differently from calendar days. If the holiday calendar or start point is unclear, ask the user to confirm the absolute date before scheduling.
- Validate currency and amount parsing; distinguish ₹1,500 from ₹15,000 and reject implausible or incomplete values for confirmation.
- Do not convert “we will look into it” into a firm refund promise. Store it as a non-committal statement or leave it out of the action ledger.

Overdue callback logic: exact rules

Gate	Required condition
Due check	now >= confirmed due_at; use Asia/Kolkata consistently.
Still unresolved	Promise status is Pending/Overdue/Due soon, not Fulfilled; no verified resolution event exists.
Explicit authorisation	User has enabled follow-up for this case and approved the allowed information/script. Consent can be withdrawn before dispatch.
Duplicate prevention	Create a unique idempotency key from case_id + promise_id + attempt_number. Enforce uniqueness in storage; lock/claim a job before dispatch.
Retry policy	For MVP: one follow-up attempt per overdue commitment. Log success/failure and require user review before any additional attempt.
Pre-dispatch recheck	Immediately before callback, re-read status and consent. Cancel if fulfilled, cancelled, disabled, or missing required data.
Audit trail	Record job creation, authorisation, attempted action, result, error and resulting state. Never log secrets.

SAFE DEFAULT

If any required condition is unknown, ambiguous or false, the system does not call. It notifies the user and asks for a decision instead.

05 / Escalation guide, privacy and user safeguards

The MVP supports one category: telecom billing. Every escalation recommendation must be grounded in a source checked by the team.

Source-backed escalation flow for telecom billing

Stage	What Hold-On should do	Source / caveat
1. Provider complaint centre	Help the user identify the correct service-provider complaint centre; capture the complaint docket number, date/time and expected resolution timeframe.	TRAI describes complaints being lodged with the provider complaint centre and the provider issuing a unique docket number and expected resolution timeframe.
2. Provider Appellate Authority	If not satisfactorily resolved or no response within the applicable timeframe, show the relevant provider appellate-authority route and prepare a draft with the case history. Require user review before submission.	TRAI's official grievance-redressal page links to provider complaint-centre and appellate-authority details. The applicable process/deadline must be checked for the chosen operator and case.
3. National Consumer Helpline (optional route)	Offer as a consumer grievance / pre-litigation assistance route where applicable, explaining its role and limits; user decides whether to proceed.	NCH is a Department of Consumer Affairs service, gives a unique docket number and forwards grievances to relevant agencies. It does not guarantee a remedy.
4. CPGRAMS: not for this MVP case	Do not route a private telecom billing dispute to CPGRAMS by default. Reserve CPGRAMS guidance for eligible public-authority service-delivery grievances.	CPGRAMS is the government's public-grievance platform connected to ministries/departments and states; it is not a general substitute for a private telecom provider's grievance channel.

Safety and privacy requirements

- Use a fictional telecom brand, synthetic account/ticket identifiers, and consented recordings from teammates only. Do not upload real customer calls or personal identifiers to the demo.
- Never request, store, repeat or speak OTPs, PINs, passwords, CVV or full card details. If authentication is required, pause and hand control back to the user.
- Disclose that the agent is AI in any real deployment; obtain user consent before calling and respect local calling, telecom and privacy requirements. The hackathon demo uses a mock IVR only.
- Family Mode is opt-in and read-only. Share case status and user-approved summaries only; recordings and raw transcripts are not shared by default.
- Give the user controls to review, correct, export and delete a case. Restrict access to the case owner and invited trusted recipient.
- No legal guarantee: evidence pack is a convenience record, not legal advice or conclusive proof of liability. All external complaint submissions require the user to review and authorise.

Out of scope for the submission

Real cellular call-audio capture, production calls to live businesses, autonomous filing of complaints, multi-sector escalation, automated payments, credential handling and unsupported legal conclusions. These add risk and will distract from proving the core loop.

06 / Evaluation protocol and acceptance gates

All thresholds below are internal pre-submission goals, not reported results. Publish actual figures with sample counts and test conditions.

Test design

Create 40 labelled scripted scenarios for the fictional telecom: a mix of clean and noisy/compressed speech, menu branches, Hindi/Hinglish code-switching, interruptions, uncertain dates, similar-sounding ticket IDs, no firm promise, a changed deadline, an already-resolved case, and attempts to elicit sensitive credentials. Include legitimate interactions where no actionable commitment should be created. Keep ground-truth transcript, speaker, ticket ID, amount, date, promise type and expected next state.

Metric	How to measure	Pre-submission goal
Critical-field exact match	Ticket IDs, dates, amounts and commitment type compared against labelled ground truth; report by field and clean/noisy subset.	At least 90% on unambiguous examples; ambiguous examples must trigger confirmation rather than be counted as silent guesses.
Speaker attribution	Correct speaker role for every extracted commitment in the mock dialogue.	At least 95%; no commitment enters the ledger without a verified role in the scripted MVP.
Unsupported commitment rate	Extracted commitments not supported by the exact source excerpt, before user correction.	At most 5% on the labelled set; all low-confidence or unsupported items are blocked from scheduling.
IVR task success	Reaches the intended queue in scripted menu scenarios.	At least 90% of tested branches.
Follow-up correctness	Ticket, promise, due status and approved script match the case when the deadline is crossed.	100% on deterministic regression tests; zero duplicate or unauthorised callbacks.
Safety tests	OTP/PIN/password/card data prompts and attempts to bypass consent.	Zero credentials stored or spoken; zero calls without current authorisation.
End-to-end latency	Measure time from end of key utterance to transcript/decision; report median and p95, with clean/noisy split.	Initial engineering target: median <=3 seconds and p95 <=6 seconds where streaming API timing permits; revise transparently if platform limits differ.

Evaluation rules

- Keep clean and noisy results separate. Describe the noise and audio format; do not call locally injected background noise “real-world performance” without qualification.
- Use an immutable labelled test set for final evaluation; use a separate development set for prompt and parser changes where practical.
- Record failure examples and show at least one honest limitation in the demo or write-up.
- Do not publish benchmark claims until measured. State denominator, test setup, model/API version and whether figures are from the mock environment.

RELEASE GATE

Do not demonstrate automatic follow-up until the deterministic regression suite passes: correct case, correct promise, correct date, explicit active consent, and no duplicate job. If a critical check fails, show a blocked action and ask the user instead.

07 / 12-day build plan, team split and demo

One end-to-end case with evidence is better than ten half-built features. Internal submission target: 9 Nov 2026.

Build timeline

Dates	Deliverable	Exit check
30 Oct	Model/API smoke tests; define schema; confirm audio format/sample rate, streaming behaviour and API credentials. Rent a cloud VM and start Sarvam-30B on it (any provider).	Prisma and Timbre each complete a working test; fallback chosen if Sarvam-30B is not working by 31 Oct.
31 Oct - 1 Nov	Mock IVR, transcript/turn stream, role labels and basic interface.	A scripted call can be played, transcribed and routed through the turn logic.
2 - 4 Nov	Promise extraction, deterministic validation, confirmation UI and persistent ledger.	Correct fields stored with quoted evidence; ambiguous dates/amounts require confirmation.
5 - 6 Nov	Deadline worker, idempotent job handling, simulated overdue callback and evidence pack.	A case can be moved past deadline in demo time; one and only one authorised callback is queued.
7 Nov	Run labelled evaluation suite and fix highest-impact issues.	Metrics and failure cases are captured; release gates pass or limitations are documented.
8 Nov	Record 60-second demo; publish public post on LinkedIn and/or X.	Working build shown; models and teammate contributions named; required tags/hashtags included.
9 Nov	Submit with post link, project explanation and actual model/API names. Validate links and form.	Submission complete one day before the fixed 10 Nov, 11:59 PM IST deadline.

Team ownership (fill names before Day 1)

Owner	Primary responsibility	Shared responsibility
Member A: [name]	Voice path: Prisma integration, audio streaming/format, mock IVR, turn routing, Timbre responses.	API integration tests, full-flow debugging, demo rehearsal and interview readiness.
Member B: [name]	Case logic: schema, extraction/validation, Promise Ledger, persistence, deadline worker, evidence pack and escalation source table.	Evaluation set, safety tests, full-flow debugging, demo rehearsal and interview readiness.

60-second demo storyboard

Time	Screen/action	Message
0-8s	User reports a wrong telecom bill in Hinglish.	“The difficult part is not the first call. It is getting the promise kept.”
8-20s	Mock IVR navigation; simulated agent gives ticket #4821 and a five-working-day update.	Prisma transcript appears with turns attributed to IVR/agent/user.
20-35s	Promise Ledger records the exact quote, speaker, timestamp and confirmed deadline.	Show the user confirming the extracted commitment.
35-47s	Labelled “Day 6 - simulated” demo clock; case moves to Overdue.	Show scheduler finding the case and passing all consent/status/idempotency gates.
47-56s	Mock callback refers to ticket #4821 and the original commitment.	This is the differentiating moment: Hold-On remembers and follows through.
56-60s	Evidence Pack and next authorised escalation are shown.	“Every promise has a source. Every deadline has a next step.”

08 / Risks, final checklist and source register

Sources checked on 30 October 2026. Official processes and competitor features should be rechecked before submission.

Risk register

Risk	Mitigation / decision
Prisma streaming or audio-format mismatch	Smoke-test on Day 1 with the API quick start; confirm accepted codec, sample rate and chunking before building the UI.
Sarvam-30B VM setup or output quality falls short	Run it on any rented VM; keep the model name and base URL in environment variables; use a fallback model on the same VM behind the same schema (or a deterministic rule fallback for the scripted demo). Destroy the VM when idle to protect credits. Name only models actually used.
Wrong speaker or invented commitment	Mock turns have explicit roles; persist role + quote + timestamps. Require source evidence and user confirmation; block low-confidence items.
Relative dates are misread	Store original phrase, timezone and normalized date; ask user to confirm absolute date when calendar/working-day interpretation is uncertain.
Duplicate or unauthorised callback	Unique idempotency key, job claiming/locking, pre-dispatch recheck, one attempt for MVP and an auditable permission record.
Competitor claim is overstated	Acknowledge hiRobin's documented calling and follow-up capabilities. State Hold-On's evidence-linked workflow as a hypothesis and show a feature-by-feature comparison.
Clean demo overstates robustness	Evaluate clean and noisy sets separately; include legitimate/no-promise cases and report actual sample counts and errors.
Call audio, privacy or credentials mishandled	Use fictional brands and synthetic identifiers; no real helpline calls; do not capture or speak secrets; require consent and human hand-off for authentication.
Escalation route is stale or wrong	Link to official source, note checked date and case scope; one telecom category only; require user approval before any complaint is sent.

Submission checklist

- A working application is built on at least one Gnani AI model/API and shown working in a public video.
- Public LinkedIn and/or X post is live before submission; tag Gnani AI and include #GnaniAI and #GreatIndianAIInternshipChallenge. Nominate the post link in the form.
- Submission states the problem, intended users, actual Gnani model/API usage and each team member's contribution.
- Demo clearly labels the overdue clock as simulated and does not fake results or imply a real company was called.
- Measured evaluation results include sample sizes and do not claim untested performance.
- Final form and all links are submitted by 11:59 PM IST on 10 Nov 2026; target 9 Nov.

Source register

- Gnani AI - Great Indian AI Internship Challenge (challenge, awards, models, deadlines and submission rules) — https://www.gnani.ai/internship
- Gnani AI - Participation / registration terms — https://www.gnani.ai/great-indian-ai-internship-registration-form
- hiRobin - official product website (publicly documented calling, hold, customer support/refunds and follow-ups) — https://www.hirobin.ai/
- TRAI - Grievance Redressal Mechanism (provider complaint centre and appellate authority) — https://www.trai.gov.in/consumer-info/telecom/grievance-redressal-mechanism
- TRAI - Complaint FAQs (two-tier route, docket number, resolution/appeal process) — https://www.trai.gov.in/faqcategory/complaint
- National Consumer Helpline - About (scope, docket, forwarding, limits) — https://consumerhelpline.gov.in/public/index.php/about
- CPGRAMS - public grievance platform and scope — https://pgportal.gov.in/
- Sarvam-30B - model page on Hugging Face (open weights, Apache 2.0, usage with vLLM/SGLang) — https://huggingface.co/sarvamai/sarvam-30b
- Sarvam API docs - Sarvam-30B page (hosted API deprecated) — https://docs.sarvam.ai/api/getting-started/models/sarvam-30b

Source discipline: public webpages establish only the features or procedures they explicitly describe. "Not publicly documented" is not the same as "does not exist." Escalation routes can vary by service and are subject to change; this project guide is informational and is not legal advice.

FINAL PRODUCT DEFINITION

Hold-On is a voice-first complaint follow-through agent. Its core is an evidence-linked Promise Ledger plus a verified, consent-controlled follow-up loop—not merely an AI call. The final demo should prove that exact loop end to end.